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Percentage

Direction (1): Study the passage given below carefully and answer the following questions.

In a school, there are total of 243 students in 5 classes (i.e. class I, II, III, IV & V). Students in Class IV are 50% more than students in Class II and students in Class III are 10 more than students in Class II. Students in Class V are 80% of students in Class IV and ratio of students in Class I to that of in Class V is 15 : 16.

1. If total students in Class – VI are equal to 150% of average number of students in Class – II & V, then find difference between total students in Class – VI and total students in Class – IV.
(A) 18 **(B)** 9
(C) 12 **(D)** 6
(E) 15

2. In an election poll, People's Party got 40% of the total valid vote. The total number of voters were 2700, out of which 30% votes were invalid. Find the amount of votes that the People's Party got.
(A) 942 **(B)** 668
(C) 567 **(D)** 756
(E) None of these

3. In a class of 80 students and 4 teachers, each student got sweets that are 25% of the total number of students, while each teacher got sweets that are 30% of the total number of students. How many sweets are there?
(A) 1696 **(B)** 1600
(C) 1969 **(D)** 1796
(E) None of these

4. Ramola's monthly income is three times Ravina's monthly income, Ravina's monthly income is fifteen percent more that Ruchika's monthly income. Ruchika's monthly income is Rs. 32,000. What is Ramola's annual income?
(A) Rs. 1,10,400 **(B)** Rs. 13,24,800
(C) Rs. 36,800 **(D)** Rs. 52,200
(E) None of these

5. Due to a 25% increase in the price of the ticket of a movie, the number of viewers are reduced to 80%. It is found that the profits of the theatre after increasing the price remained the same. What is the initial number of viewers and the initial rate of tickets?
(A) 150, Rs. 30
(B) 120, Rs. 20
(C) 200, Rs. 40
(D) Cannot be determined
(E) None of the above

6. The price of a one room flat is Rs. 5,15,000. It was insured to 80% of its price. The flat was burnt completely in an accident and the insurance company paid 90% of the insurance. What was the difference between the price of the flat and the amount received?
(A) Rs. 240800 **(B)** Rs. 362600
(C) Rs. 144200 **(D)** Rs. 447800
(E) None of these

Direction (7 – 11): Read the data carefully and answer the questions.

There are total 2100 students in two schools A and B and there are three streams in each school i.e. Art, Science & Commerce. 1/3rd

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of total students in school A are in Commerce stream and 100/6% of total students in school B are in Science stream. Sum of total students in Commerce stream in A and Science stream in B is 500. 40% of total students in school B are in Commerce stream and 40% of total students in school A are in Art stream.

7. Find the ratio of total students in Commerce stream in school B to total students in Science stream in school A?
(A) 2 : 1 **(B)** 3 : 2
(C) 2 : 3 **(D)** 4 : 3
(E) 1 : 2

8. Find the average number of students in Arts stream in school A and B?
(A) 330 **(B)** 420
(C) 440 **(D)** 380
(E) 450

9. Total students in Art stream in school B is what percent more than total students in Science stream in school A?
(A) 106.67% **(B)** 108.33%
(C) 87.67% **(D)** 116.67%
(E) 122.34%

10. Out of total students in Art stream of schools A and B, ratio of boys and girls is 5 : 3 and 7 : 6 respectively. Find the difference between boys and girls in Art stream of schools A and B together?
(A) 140 **(B)** 120
(C) 110 **(D)** 150
(E) 130

11. In school C total students are 1050 and students in Science stream of school C are 25% more than total students in Commerce stream in school B, then find total students of Art & Commerce stream in school C is how much less than total students in Art and Commerce stream in school A?
(A) 33.5% **(B)** 31.8%
(C) 32.7% **(D)** 35.3%
(E) 30.8%

Direction (12 – 16): In the table, data regarding total number of employees and total number of employees with 5+ years of experience in four companies – A, B, C and D is given. Read the data carefully & answer the following questions.

Company	Total number of employees	Total number of employees with 5+ years of experience
A	240	96
B	280	84
C	320	160
D	360	162

Note: Total number of employees = Number of employees with 5+ years of experience + Number of employees with less than 5 years of experience

12. Find the ratio of number of employees with less than 5 years of experience in company A and total number of employees with 5+ years of experience in B & C together.
(A) 35 : 62 **(B)** 37 : 61
(C) 36 : 61 **(D)** 35 : 66
(E) 36 : 59

13. Total number of employees with less than 5 years of experience in companies C and D together is what percent of total employees in company D?
(A) 98.8% **(B)** 99.8%
(C) 97.6% **(D)** 99.4%
(E) None of these

14. Number of males with less than 5 years of experience in company A is 70 and number of male with 5+ years of experience in company C is 48. What is the total number of females with less than 5 years of

experience in company A and with 5+ years of experience in company C?
(A) 186 **(B)** 172
(C) 164 **(D)** 196
(E) 204

15. What is the difference between average number of employees with less than 5 years of experience in company A and C together and the average number of emloyees with 5+ years of experience in company B and D together?
(A) 23 **(B)** 33
(C) 27 **(D)** 31
(E) 29

16. If number of females with less than 5 years of experience in company D is 98 and number of females with 5+ years of experience is 40% of number of males with less than 5 years of experience in same company, what is the total number of male with 5+ years of experience in company D?
(A) 120 **(B)** 134
(C) 122 **(D)** 118
(E) 112

Direction (17-20): Read the following passage and answer the following questions.

As sum of Rs. 90,000 is to be distributed among 5 person i.e. A, B, C, D and E respectively. Ratio of share of A to share of D is 6 : 5. Difference in share of D and share of C is $11\frac{1}{9}\%$ of total sum of all. Share of D is Rs. 3000 more than share of E. Share of B is $11\frac{1}{9}\%$ more than share of A. Share of C is more than share of D.

17. The population of a town is 4320 and ratio between men and women is 4 : 5. If there are 60% literate among men and 50% literate among women. Find the total number of illiterate men is what percent of the total number of literate women in town?
(A) 36% **(B)** 64%
(C) 72% **(D)** 78%
(E) 56%

18. A man deposit 10% of his salary in PF. He saves 30% of the remaining. The ratio of his expense on medicine and groceries is 3 :4 of the remaining salary after saving. If his expense on the medicine was Rs 8100. Find the monthly salary.
(A) Rs 30,000 **(B)** Rs 25,000
(C) Rs 20,000 **(D)** Rs 35,000
(E) None of these

19. In an examination A got 25% marks more than B, B got 10% less than C and C got 25% more than D. If D got 320 marks out of 500, the marks obtained by A were.
(A) 405 **(B)** 450
(C) 360 **(D)** 400
(E) 480

20. A man deposited 14% of the initial amount to his locker. And again after some time he deposited 45% of the increased amount. Now the amount becomes Rs 16530. How much was the initial amount?
(A) Rs 10500 **(B)** Rs 10000
(C) Rs 9500 **(D)** Rs 9000
(E) None of these

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Percentage

Solution

Answer:(1):

Let student in Class – II be 2x.

So, student in Class – IV = $\frac{150}{100} \times 2x$

$= 3x$

Student in class – III = $(10 + 2x)$

Student in class – V = $\frac{80}{100} \times 3x$

$= 2.4x$

Student in class – I = $2.4x \times \frac{15}{16}$

$= 2.25x$

ATQ,

$2.25x = 2x = (10 + 2x) + 3x + 2.4x = 243$

$11.65x = 233$

$x = 20$

Class	Total student
I	**45**
II	**40**
III	**50**
IV	**60**
V	48

1. Answer: (D)

Total students in Class – VI = $\frac{150}{100} \times \left(\frac{40 + 48}{2}\right)$

$= 66$

Required difference = $66 - 60$

$= 6$

2. Answer: (D)

Number of invalid votes = 30% of 2700

= 810.

Thus valid vote = (2700 -810) = 1890

Valid votes polled by People's Party = 40% of 1890 = 756.

3. Answer: (A)

Each student gets sweets which is equal to the value of 25% of the total number of students.

This can be calculated as (25/100) × 80 = 20 sweets

Similarly, each teacher gets (30/100) × 80 = 24 sweets

Total number of sweets = number of students × sweets per student + number of teachers × sweets per teacher = 80 × 20 + 4 × 24 = 1696

4. Answer: (B)

Ramola = Ravina × 3

Ravina = Ruchira × (115/100)

= 32000 × (115/100)

= Rs. 36800

∴ Ramola's annual income = 36800 × 3 × 12

= Rs. 13,24,800

Hence, option 2 is correct.

5. Answer: (D)

Let in initial price of the ticket be Rs. a and the initial number of viewers be b.

Total revenue = ab

Now the price is changed to 1.25a and the number of viewers are changed to 0.8 b.

Total revenue = 1.25a × 0.8b = ab

This means whatever ne the price and number of viewers, 25% increase in price and 20% decrease in viewers will not change the profit.

Hence, data is insufficient.

6. Answer: (C)

Price of the flat = Rs.5,15,000

Flat was insured to 80% of its price

Insured price = 515000 × 80/100

Insurance company paid 90% of the insurance.

Amount paid by insurance company

= 515000 × 80/100 × 90/100 = 370800.

Difference between the price of the flat and the amount received

= Rs. (5,15,000 - 370800) = Rs. 144200

Answer: (7 – 11):

The correct answer is Option 1 i.e. 2 : 1

Suppose Total students in schools A and B are 'a' and 'b' respectively.

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Students in Commerce stream in school A = a/3
Students in Science stream in school B = 100/6 × 1/100 × b = b/6
Since Sum of total students in Commerce stream in A and Science stream in B is 500:
a/3 + b/6 = 500
2a + b = 3000 ..... (1) Since there are total 2100 students in two schools A and B:
a + b = 2100 ..... (2) From equation 1 and 2:
a = 900 & b = 1200
So,
Students in Commerce stream in school A = a/3 = 300
Students in Science stream in school B = b/6 = 400
40% of total students in school B are in Commerce stream and 40% of total students in school A are in Art stream.
So,
Students in Commerce stream in school B = 1200 × 0.4 = 480
Students in Science stream in school A = 900 × 0.4 = 360
Now,
Students in Art stream in school A = 900 – 300 – 360 = 240
Students in Art stream in school B = 1200 – 480 – 200 = 520
Hence, we can prepare the following table:

Streams	A	B
Art	360	520
Commerce	300	480
Science	240	200

7. Answer:(C)

8. Answer: (C):
Hence, average number of students in Arts stream in school A and B = (360 + 520)/2 = 440

9. Answer: (D):
Total students in Art stream in school B = 520
Total students in Science stream in school A = 240
Hence,
Required percent = [(520 – 240)/240] × 100 = 116.67%

10. Answer: (E):
School A:
Total students in Art stream = 360
Number of boys = 360 × 5/8 = 225
Number of girls = 360 × 3/8 = 135
School B:
Total students in Art stream = 520
Number of boys = 520 × 7/13 = 280
Number of girls = 520 × 6/13 = 240
Now,
Number of boys in Art stream of schools A and B together = 225 + 280 = 505
Number of girls in Art stream of schools A and B together = 135 + 240 = 375
Hence,
Required difference = 505 – 375 = 130

11. Answer: (B):
Since students in Science stream of school C are 25% more than total students in Commerce stream in school B;
Students in Science stream of school C = 1.25 × 480 = 600
Given: In school C total students are 1050.
Hence,
Total students of Art & Commerce stream in school C = 1050 – 600 = 450
Total students in Art and Commerce stream in school A = 360 + 300 = 660
Hence,
Required percentage = [(660 – 450)/660] × 100 = 31.8%

12. Answer: (C):
The correct answer is Option 3 i.e. 36 : 61
From the table:
Number of employees with less than 5 years of experience in company A = 240 – 96 = 144
Total number of employees with 5 + years of experience in B & C together = 84 + 160 = 244
Hence, ratio = 144 : 244 = 36 : 61

13. Answer: (D):

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The correct answer is Option 4 i.e. 99.4%
From the table:
Total number of employees with less than 5 years of experience in companies C and D together
= (320 – 160) + (360 – 162) = 160 + 198
= 358
Total employees in company D = 360
Hence,
Required percentage = 358/360 × 100
= 99.4%

14. Answer: (A):
The correct answer is Option 1 i.e. 186
From the table:
Total number of employees with less than 5 years of experience in company A = 240 – 96 = 144
Given: Number of males with less than 5 years of experience in company A is 70
So,
Total number of females with less than 5 years of experience in company A = 144 – 70 = 74
And
Total number of employees with 5 + years of experience in company C = 160
Given: Number of males with 5 + years of experience in company C is 48
So,
Total number of females with 5 + years of experience in company C = 160 – 48 = 112
Hence, sum = 74 + 112 = 186

15. Answer: (E):
The correct answer is Option 5 i.e. 29
From the table:
Number of employees with less than 5 years of experience in company A = 240 – 96 = 144
Number of employees with less than 5 years of experience in company C = 320 – 160 = 160
Hence, Average = (144 + 160)/2 = 152
Number of employees with 5 + years of experience in company B = 84
Number of employees with 5 + years of experience in company D = 162
Hence, Average = (84 + 162)/2 = 123
Hence, difference = 152 – 123 = 29

16. Answer: (C):
The correct answer is Option 3 i.e. 122
From the table:
Number of employees with less than 5 years of experience in compnay D = 360 – 162 = 198
Number of females with less than 5 years of experience in company D is 98
So, number of males with less than 5 years of experience in company D = 198 – 98 = 100
Given: number of females with 5 + years of experience is 40% of number of males with less than 5 years of experience in company D.
So,
Number of females with 5 + years of experience in company D = 0.4 × 100 = 40
Hence, total number of male with 5 + years of experience in company D = 162 – 40 = 122

17. Answer: (B)
Illiterate Men = $\frac{4}{9} \times 4320 \times \frac{40}{100} = 768$
Literate Woman
= $\frac{5}{9} \times 4320 \times \frac{50}{100} = 1200$
∴ Required percentage
= $\frac{768}{1200} \times 100 = 64\%$

18. Answer: (A)
Let monthly salary = Rs. 100x
ATQ,
$100x \times \frac{90}{100} \times \frac{70}{100} = 63x$
Let total expense on Medical and groceries
= 3y + 4y = 7y
So, 7y = 63*x*
y = 9x
Given, 3y = 8100,
So, *y* = 2700
Now, *x* = 300
∴ monthly salary of the man

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= 100x = Rs. 30,000

19. **Answer: (B)**

Marks obtained by D = 320

Marks obtained by C

$= 320 \times \frac{125}{100} = 400$

Marks obtained by B

$= 400 \times \frac{(100-10)}{100} = 360$

Marks obtained by A

$= 360 \times \frac{125}{100} = 450$

Hence, required marks obtained by A = 450

20. **Answer: (B)**

Let initial amount = 100

Now, total amount

$= 100 + 14 + \frac{45}{100} \times 114$

$= 165.3$

$\therefore 165.3 \rightarrow 16530$

$1 \rightarrow \frac{16530}{165.3}$

$\therefore 100 \rightarrow 100 \times 100 = 10{,}000\ Rs.$